# Core 90 - Union-find and connectivity

3 questions to study here, each on its own page, with the full interview answer and code

Nothing to skip: every question that fits this topic is covered here

| 2 | 1 |
|---|---|
| MEDIUM | HARD |

**How to use**

- Questions run from Easy to Medium to Hard, newest last-asked date first inside each level.
- PDFs run from the one with the most questions to the fewest. A question has pages in only one PDF; the others list it as SKIP
- Every page has < Prev | Contents | Next > at the bottom. Tap a title in the Contents to jump to it.
- Select code and press Cmd/Ctrl+C to copy it.

Go to Contents >

# Contents

**Core 90 - Union-find and connectivity | 3 questions | tap a title to jump to it**


| | | | |
|---|---|---|---|
| M | 1. Number of Islands | 1Point3Acres | 2026-09-30 |
| M | 2. 2D Points Connected Components by Radius | 1Point3Acres | 2026-08-04 |
| H | 3. Friend Graph with Timeline (Friend / Unfriend Events) | 1Point3Acres | 2026-09-29 |

**STUDY PATH: question 53 of 167 (Core 90)**

Priority score 9.59 | Union-find and connectivity | asked 20 days ago | frequency Medium

**MEDIUM | IN APP LC 200 | 1Point3Acres | onsite-coding, phone-screen | 1 of 3**

**★ TOP 75: rank 17 in the Medium and Hard priority list**

# ★ Number of Islands

**ASKED 2026-09-30 | 20 DAYS AGO | LAST 30 DAYS**

**STUDY: TYPE** | **ALL TIME 64.8** | **30 DAYS 33** | **PRIORITY 19/228**

## ABOUT THIS QUESTION

**Topic:** Core 90 - Union-find and connectivity

**Optimal answer from:** your app's interview approach (LeetCode #200)

**LeetCode match:** LC 200 Number of Islands

Frequency: Medium | Last asked: 2026-09-30 | Roles: SWE, MLE

**Original:** https://www.1point3acres.com/interview/problems/company/google/number-of-islands

## QUESTION

EXAMPLES

```
[[1,1,0,0],[1,1,0,0],[0,0,1,1],[0,0,1,1]] -> 2.
Streaming: [[0,0],[0,1],[1,1]] adds -> counts [1, 1,
    1] (last addition merges the two land cells).
```

## APPROACH

Write both the static count and the streaming addLand version from memory in under 15 min total.
Be ready to switch between DFS / BFS / Union-Find on demand.

## HOW TO THINK ABOUT IT

Each island is a connected group of land cells, so count how many times you start a fresh search at an unvisited land cell. The search itself (DFS or BFS) just marks everything reachable. If land is added over time, do not rerun the search: Union-Find keeps the count up to date.

**Basis:** LeetCode 200, Number of Islands, and LeetCode 305, Number of Islands II (streaming additions with Union-Find).

## BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

## BRUTE FORCE

```
def brute(grid):
    seen, count = set(), 0
    for r, c in all_cells(grid):
        if grid[r][c] == 0 or (r, c) in seen:
            continue
        count += 1
        dfs(r, c)  # marks the whole island
    return count
def dfs(r, c):
    if out_of_grid(r, c) or grid[r][c] == 0:
        return
    if (r, c) in seen:
        return
    seen.add((r, c))
    for dr, dc in DIRECTIONS:
        dfs(r + dr, c + dc)
```

| TIME | SPACE |
|---|---|
| O(m * n) | O(m * n) recursion |

WHY IT IS SLOW

Time is already O(m * n); the weakness is the recursion depth, which can reach m * n on a snake-shaped island and overflow the stack. An explicit queue or stack avoids that.

## OPTIMAL SOLUTION · Iterative DFS flood fill

| TIME | SPACE |
|---|---|
| O(mn) | O(mn) |

**Code:** see FULL INTERVIEW CODE below (the optimal solution is its main function).

## FOLLOW-UP 1 What if land can also be removed?

Union-Find cannot split sets. Either rebuild the affected component with a local BFS from each former neighbour, or process the whole timeline offline with a segment tree over time and union-find with rollback (the offline dynamic connectivity technique).

## FOLLOW-UP 2 How do you handle a grid too large for memory?

Process it in horizontal strips: label islands within the strip, keep only the boundary row's labels, and merge labels across strips with a union-find over label ids. Memory is O(width + number of labels on the boundary).

## FOLLOW-UP 3 Find the area of the largest island.

Count the cells during each search and keep the maximum. Same traversal, one counter.

**FOLLOW-UP 4 The grid is huge and does not fit in memory.**

Process it in strips of rows, label islands in each strip, and merge labels across strip boundaries with a Union-Find over the labels on the boundary row. **Cost:** O(m * n) time overall and O(width + labels on the boundary row) memory per strip.

## FULL INTERVIEW CODE · base question + every follow-up

**One program for the whole question:** the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
# MAIN QUESTION: Number of Islands
#
# Examples: [[1,1,0,0],[1,1,0,0],[0,0,1,1],[0,0,1,1]] -> 2
.
# Streaming: [[0,0],[0,1],[1,1]] adds -> counts [1, 1, 1]
# (last addition merges the two land cells).
#
# FOLLOW-UPS (answered below, in this order):
# 1. What if land can also be removed?
# 2. How do you handle a grid too large for memory?
# 3. Find the area of the largest island.
# 4. The grid is huge and does not fit in memory.
#
# INTERVIEW NOTES
# Each island is a connected group of land cells, so count
# how many times you start a fresh search at an unvisited
# land cell. The search itself (DFS or BFS) just marks
# everything reachable. If land is added over time, do not
# rerun the search: Union-Find keeps the count up to date.
#
# BRUTE FORCE (say it first): Recursive flood fill
# Scan; at each unvisited land cell count one island and
# recurse in four directions to mark it. The textbook
# answer and the oracle; deep recursion can overflow on a
# large snake-shaped island. Time O(m * n). Space O(m * n)
# recursion.


# BRUTE FORCE (say it first): Recursive flood fill | time
# O(m * n)
def islands_flood(grid):
    if not grid:
        return 0
    m, n = len(grid), len(grid[0])
    seen = set()

    def fill(r, c):
        if (
            not (0 <= r < m and 0 <= c < n)
            or grid[r][c] != 1
            or (r, c) in seen
        ):
            return
        seen.add((r, c))
        for dr, dc in (
            (1, 0),
            (-1, 0),
            (0, 1),
            (0, -1),
        ):
            fill(r + dr, c + dc)

    count = 0
    for r in range(m):
        for c in range(n):
            if grid[r][c] == 1 and (r, c) not in seen:
                count += 1
                fill(r, c)
    return count


# HELPERS for the base question
DIRS = ((1, 0), (-1, 0), (0, 1), (0, -1))


def neighbors(r, c, rows, cols):
    # the cells beside (r, c) that are inside the grid
    for dr, dc in DIRS:
        nr, nc = r + dr, c + dc
        if 0 <= nr < rows and 0 <= nc < cols:
            yield nr, nc


def flood(grid, start, seen):
    # visit all land joined to start; return the count
    rows, cols = len(grid), len(grid[0])
    seen.add(start)
    stack, size = [start], 0
    # a stack instead of recursion: no recursion limit
    while stack:
        r, c = stack.pop()
        size += 1
        for nb in neighbors(r, c, rows, cols):
            if grid[nb[0]][nb[1]] == 1 and nb not in seen:
                seen.add(nb)
                stack.append(nb)
    return size


def island_sizes(grid):
    if not grid:
        return []
    seen, sizes = set(), []
    for r in range(len(grid)):
        for c in range(len(grid[0])):
            # an island starts at land not visited yet
            if grid[r][c] == 1 and (r, c) not in seen:
                sizes.append(flood(grid, (r, c), seen))
    return sizes


# BASE QUESTION (optimal): Iterative DFS flood fill
# Idea: Scan every cell; each unvisited land cell starts a
# new island, and an explicit stack marks all connected
# land as visited. Iterative DFS avoids Python's recursion
# limit on large grids.
def num_islands(grid):
    # one entry per island
    return len(island_sizes(grid))


# VARIANT: land cells arrive one at a time
class IslandCounter:
    """Follow-up: land cells arrive one at a time."""

    def __init__(self, rows, cols):
        self.rows, self.cols = rows, cols
        # land cell -> parent cell (union-find)
        self.parent = {}
        self.count = 0

    def find(self, x):
        # path halving keeps the trees flat
        while self.parent[x] != x:
            self.parent[x] = self.parent[self.parent[x]]
            x = self.parent[x]
        return x
```

```
    def add_land(self, r, c):
        # adding the same cell twice changes nothing
        if (r, c) in self.parent:
            return self.count
        self.parent[(r, c)] = (r, c)
        self.count += 1
        # join with every land neighbour
        for nb in neighbors(r, c, self.rows, self.cols):
```

```
            if nb in self.parent:
                a, b = self.find((r, c)), self.find(nb)
                if a != b:
                    self.parent[a] = b
                    # two islands merged into one
                    self.count -= 1
        return self.count


# FOLLOW-UP 3: Find the area of the largest island.
# Answer: Count the cells during each search and keep the
# maximum. Same traversal, one counter.
def largest_island(grid):
    # follow-up: the area of the biggest island
    return max(island_sizes(grid), default=0)


if __name__ == "__main__":
    g = [[1, 1, 0, 0], [1, 1, 0, 0],
         [0, 0, 1, 1], [0, 0, 1, 1]]
    assert num_islands(g) == 2
    assert largest_island(g) == 4
    assert num_islands([]) == 0
    # streaming adds: counts stay 1, 1, 1
    ic = IslandCounter(2, 2)
    adds = [(0, 0), (0, 1), (1, 1)]
    counts = [ic.add_land(r, c) for r, c in adds]
    assert counts == [1, 1, 1]
    # two separate cells make two islands
    ic = IslandCounter(3, 3)
    assert ic.add_land(0, 0) == 1
    assert ic.add_land(2, 2) == 2
    assert ic.add_land(1, 1) == 3
```

## NOTES FROM CANDIDATES

For streaming, Union-Find with path compression is the natural choice — each addLand is amortized alpha(n).
In one onsite, the follow-up arrived after a working BFS solution and explicitly required a Union-Find rewrite; retrofitting the original traversal consumed the remaining time.
Common edge cases: diagonal connectivity (some prompts use 8-connectivity), grid with no land.

## PREPARATION

Write both the static count and the streaming addLand version from memory in under 15 min total.
Be ready to switch between DFS / BFS / Union-Find on demand.

**STUDY PATH: question 54 of 167 (Core 90)**

Priority score 5.63 | Union-find and connectivity | asked 77 days ago | frequency Low

MEDIUM | 1Point3Acres | phone-screen | 2 of 3

# 2D Points Connected Components by Radius

ASKED 2026-08-04 | 77 DAYS AGO | LAST 90 DAYS

STUDY: TALK | ALL TIME 40.7 | 30 DAYS 0 | PRIORITY 91/228

## ABOUT THIS QUESTION

**Topic:** Core 90 - Union-find and connectivity

**Optimal answer from:** written and tested here for this exact question

**Related in your app (a neighbouring problem, not this exact question):** LeetCode #547 Number of Provinces

**LeetCode match:** LC 547 Number of Provinces (related)

Frequency: Low | Last asked: 2026-08-04 | Roles: SWE

**Original:** https://www.1point3acres.com/interview/problems/company/google/points-connected-components-by-radius

## QUESTION

Given 2D points and radius **r**, connect two points when their distance is at most **r**; connectivity is transitive, and the output is the final number of clusters. The interviewer accepted or preferred a brute-force neighbor scan after discussing optimization ideas.

**Input:** a list of 2D points and a radius r.
Two points are connected if their Euclidean distance is <= r.
Connectivity is transitive: if A connects to B and B connects to C, all three belong to the same cluster.
Return the number of final clusters.

## APPROACH

Implement connected components once with DFS/BFS and once with Union-Find.
Practice stating the O(n^2) brute-force baseline clearly before proposing spatial indexing.
Dry-run a case where three points form one cluster only through transitive links.

## HOW TO THINK ABOUT IT

Two points are neighbours when their squared distance is at most r squared (no square roots needed), and connectivity is transitive, so this is counting connected components: union-find or a DFS over the neighbour relation. Checking every pair is O(n^2); bucketing points into cells of side r means a point only needs to be compared with the 3 x 3 cells around it.

**Basis:** LeetCode 547, Number of Provinces, where two points are linked when their distance is at most r.

## BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
def brute(points, r):
    graph = defaultdict(list)
    ids = range(len(points))
    for i, j in combinations(ids, 2):
        if dist2(points[i], points[j]) <= r * r:
            graph[i].append(j)
            graph[j].append(i)
    seen, clusters = set(), 0
    for i in range(len(points)):
        if i not in seen:
            clusters += 1
            dfs(graph, i, seen)
    return clusters
```

| TIME | SPACE |
|---|---|
| O(n^2) | O(n) |

WHY IT IS SLOW

Every pair is compared, O(n^2). Putting points in cells of side r means each point is compared only with the 3 x 3 cells around it.

## OPTIMAL SOLUTION · Union-find with a spatial grid

| TIME | SPACE |
|---|---|
| O(n) about when points are spread out (O(n^2) if they all share a cell) | O(n) |

**Code:** see FULL INTERVIEW CODE below (the optimal solution is its main function).

## FOLLOW-UP 1 Why compare squared distances?

sqrt is slow and inexact; dist <= r is the same as dist^2 <= r^2 for non-negative numbers, and with integer coordinates the comparison is exact. It also avoids floating-point surprises right at the boundary.

## FULL INTERVIEW CODE · base question + every follow-up

**One program for the whole question:** the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
# MAIN QUESTION: 2D Points Connected Components by Radius
#
# Given 2D points and radius r, connect two points when
# their distance is at most r; connectivity is transitive,
# and the output is the final number of clusters. The
# interviewer accepted or preferred a brute-force neighbor
# scan after discussing optimization ideas. Input: a list
# of 2D points and a radius r. Two points are connected if
# their Euclidean distance is <= r. Connectivity is
# transitive: if A connects to B and B connects to C, all
# three belong to the same cluster. Return the number of
# final clusters.
#
# FOLLOW-UPS (answered below, in this order):
```

```python
# 1. Why compare squared distances?
#
# INTERVIEW NOTES
# Two points are neighbours when their squared distance is
# at most r squared (no square roots needed), and
# connectivity is transitive, so this is counting
# connected components: union-find or a DFS over the
# neighbour relation. Checking every pair is O(n^2);
# bucketing points into cells of side r means a point only
# needs to be compared with the 3 x 3 cells around it.
#
# BRUTE FORCE (say it first): Compare every pair and run a
# DFS
# Compare every pair of points, build the neighbour lists
# and count components with a DFS. This is the baseline
# the interviewer accepted, and it is fine for a few
# thousand points. Time O(n^2). Space O(n).


from collections import defaultdict


# BRUTE FORCE (say it first): Compare every pair and run a
# DFS | time O(n^2)
def components_brute(points, r):
    n, seen, count = len(points), set(), 0
    near = (
        lambda a, b: (a[0] - b[0]) ** 2
        + (a[1] - b[1]) ** 2
        <= r * r
    )
    for s in range(n):
        if s in seen:
            continue
        count += 1
        stack = [s]
        seen.add(s)
        while stack:
            u = stack.pop()
            for v in range(n):
                if v not in seen and near(
                    points[u], points[v]
                ):
                    seen.add(v)
                    stack.append(v)
    return count


# HELPERS for the base question
class UnionFind:
    def __init__(self, n):
        self.parent = list(range(n))

    def find(self, x):
        # path halving keeps the trees flat
        while self.parent[x] != x:
            self.parent[x] = self.parent[self.parent[x]]
            x = self.parent[x]
        return x

    def union(self, a, b):
        self.parent[self.find(a)] = self.find(b)


# FOLLOW-UP 1: Why compare squared distances?
# Answer: sqrt is slow and inexact; dist <= r is the same
# as dist^2 <= r^2 for non-negative numbers, and with
# integer coordinates the comparison is exact. It also
# avoids floating-point surprises right at the boundary.
# NEEDED BY THE BASE QUESTION: it is also the answer to
# the follow-up below
def near(p, q, r):
    # squared distance: no square root, exact for ints
    dx, dy = p[0] - q[0], p[1] - q[1]
    return dx * dx + dy * dy <= r * r


# BASELINE: compare every pair
def count_clusters_pairs(points, r):
    # the accepted baseline: compare every pair, O(n^2)
    uf = UnionFind(len(points))
    for i in range(len(points)):
        for j in range(i + 1, len(points)):
            if near(points[i], points[j], r):
                uf.union(i, j)
    return len({uf.find(i) for i in range(len(points))})


# BASE QUESTION (optimal): Union-find with a spatial grid
# Idea: Put each point in the cell (x // r, y // r). Two
# points within distance r are in the same or adjacent
# cells, so only those 9 cells are compared.
def count_clusters(points, r):
    # grid of cells r wide: only 9 cells need a look
    size = r if r > 0 else 1
    cells = defaultdict(list)
    for i, (x, y) in enumerate(points):
        cells[(x // size, y // size)].append(i)
    uf = UnionFind(len(points))
    for i, (x, y) in enumerate(points):
        cx, cy = x // size, y // size
        for dx in (-1, 0, 1):
            for dy in (-1, 0, 1):
                cell = cells.get((cx + dx, cy + dy), ())
                for j in cell:
                    q = points[j]
                    if j > i and near((x, y), q, r):
                        uf.union(i, j)
    return len({uf.find(i) for i in range(len(points))})


if __name__ == "__main__":
    # three points joined only through the middle one
    line = [(0, 0), (3, 0), (6, 0)]
    assert count_clusters(line, 3) == 1
    assert count_clusters_pairs(line, 3) == 1
    # the far point stays alone
    far = [(0, 0), (3, 0), (7, 0)]
    assert count_clusters(far, 3) == 2
    # radius 0 only joins identical points
    same = [(1, 1), (1, 1), (2, 2)]
    assert count_clusters(same, 0) == 2
    assert count_clusters([], 5) == 0
    # both methods agree on a scatter of points
    pts = [(i * 7 % 13, i * 5 % 11) for i in range(30)]
    for r in (0, 1, 2, 3, 5):
        assert count_clusters(pts, r) == (
            count_clusters_pairs(pts, r))
```

## NOTES FROM CANDIDATES

DFS, BFS, or Union-Find all fit the core connected-components model.
For many points, discuss how to avoid checking every pair by narrowing neighbor candidates spatially; in this round, the interviewer still preferred the brute-force implementation path.
Clarify squared-distance comparison to avoid unnecessary square roots.

## PREPARATION

Implement connected components once with DFS/BFS and once with Union-Find.
Practice stating the O(n^2) brute-force baseline clearly before proposing spatial indexing.
Dry-run a case where three points form one cluster only through transitive links.

**STUDY PATH: question 55 of 167 (Core 90)**

Priority score 6.5 | Union-find and connectivity | asked 21 days ago | frequency Medium

**HARD | 1Point3Acres | onsite-coding, phone-screen | 3 of 3**

# Friend Graph with Timeline (Friend / Unfriend Events)

**ASKED 2026-09-29 | 21 DAYS AGO | LAST 30 DAYS**

**STUDY: TALK** | **NO LEETCODE FREQUENCY** | **PRIORITY 186/228**

## ABOUT THIS QUESTION

**Topic:** Core 90 - Union-find and connectivity

**Optimal answer from:** written and tested here for this exact question

Frequency: Medium | Last asked: 2026-09-29 | Roles: SWE

**Original:** https://www.1point3acres.com/interview/problems/company/google/friend-graph-with-timeline

## QUESTION

### EXAMPLES

```
Base: N=3, events [(1,A,B,friend), (5,B,C,friend)] ->
    answer 5.
Follow-up 1: after the above events, query (0, A, C)
    -> false, (5, A, C) -> true.
```

## APPROACH

Have weighted/path-compressed union-find ready and warm. Practice the offline trick: segment tree over time intervals containing each edge, DFS the tree applying / rolling back unions.

Memorize the punchline for fully-dynamic: "online deletion breaks union-find; offline is OK; for online you need Holm–Lichtenberg or link-cut tree (O(polylog n))."

## INTERVIEW-DAY PLAN

### CLARIFY FIRST

Ask what the answer is for the base question (earliest time all N people are connected), what to return if it never happens, whether unfriend events exist, and whether queries ask about a past time.

### BRUTE FORCE TO SAY OUT LOUD

For each query rebuild the graph from the events up to that time and run BFS. O(E + N) per query.

### OPTIMAL IDEA

Base: sort events and union-find, return the time the component count hits 1. Query at a time t: union by size without path compression and record each union's time; two people are connected at t if the largest time on the path between them is at most t. Unfriend: offline, place each edge's lifetime into a segment tree over time and run DFS with a rollback union-find, checking connectivity at the leaves.

### DRY-RUN THESE CASES

## INTERVIEW-DAY PLAN

The example giving 5. A graph that never connects. Two events at the same time. Unfriend then refriend.

### LIKELY FOLLOW-UPS

Online unfriend (Holm-de Lichtenberg-Thorup or link-cut trees), counting components over time.

### PITFALLS

Using path compression where rollback is needed. Treating unfriend like union-find undo.

## HOW TO THINK ABOUT IT

Everyone becomes connected at the first moment the graph has one component. If friendships only appear, sort by time and use union-find (components only merge). Unfriending breaks that: removing an edge can split a component and union-find cannot undo. With all events known up front, handle each edge's lifetime in a segment tree over time and use a rollback union-find.

**Basis:** LeetCode 1101, The Earliest Moment When Everyone Become Friends, plus connectivity queries over time and unfriend events (offline dynamic connectivity). Assumption: events are (time, a, b, 'friend'|'unfriend').

## BRUTE FORCE

### BRUTE FORCE · PSEUDO-CODE

```
def brute(n, events):
    for t in sorted({e.time for e in events}):
        graph = build_graph(events, upto=t)
        if dfs_count(graph, start=0) == n:
            return t  # everyone reached
    return -1
```

| TIME | SPACE |
|---|---|
| O(T * (N + E)) | O(N + E) |
| for T distinct times | |

### WHY IT IS SLOW

Each time point replays all events and searches the whole graph, O(T * (N + E)). A union-find over sorted events (friend-only) is near-linear; unfriend events need offline dynamic connectivity.

## OPTIMAL SOLUTION · Base: earliest time everyone is connected (Union-Find on sorted events)

| TIME | SPACE |
|---|---|
| O(E log E) | O(N) |

**Code:** see FULL INTERVIEW CODE below (the optimal solution is its main function).

**FOLLOW-UP 1 Why does unfriending break plain Union-Find?**

A union merges two sets and forgets which edge caused it, so removing that edge cannot tell which members should separate. Online, fully dynamic connectivity needs heavy structures (Holm-de Lichtenberg-Thorup or link-cut trees, polylogarithmic time). When all events are known in advance, the offline segment-tree-over-time plus rollback DSU is the practical answer and the one to code in an interview.

**FOLLOW-UP 2 What is the simple answer if the interviewer wants something quick?**

Rebuild the graph from the events up to time t and run BFS or DFS for each query: O(E + N) per query. State the cost and then explain how the offline method amortizes it. A correct simple answer is accepted; naming the optimal one shows depth.

**FOLLOW-UP 3 Report how many friend groups exist after each event (friends only).**

Union-find with a component counter: decrement when a union merges two different roots; record the counter after each event. O(E alpha(N)).

**FOLLOW-UP 4 Queries ask 'were A and B connected at time t?' with no unfriending.**

Union by size without path compression, recording the time of each union on the edge it creates; two people are connected at t if the largest time on the path between them is at most t.

**FULL INTERVIEW CODE · base question + every follow-up**

**One program for the whole question:** the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
# MAIN QUESTION: Friend Graph with Timeline (Friend /
# Unfriend Events)
#
# Examples: Base: N=3, events [(1,A,B,friend),
# (5,B,C,friend)] -> answer 5. Follow-up 1: after the abov
e
# events, query (0, A, C) -> false, (5, A, C) -> true.
#
# FOLLOW-UPS (answered below, in this order):
# 1. Why does unfriending break plain Union-Find?
# 2. What is the simple answer if the interviewer wants
# something quick?
# 3. Report how many friend groups exist after each event
# (friends only).
# 4. Queries ask 'were A and B connected at time t?' with
# no unfriending.
#
# INTERVIEW NOTES
# Everyone becomes connected at the first moment the graph
# has one component. If friendships only appear, sort by
# time and use union-find (components only merge).
# Unfriending breaks that: removing an edge can split a
# component and union-find cannot undo. With all events
# known up front, handle each edge's lifetime in a segment
# tree over time and use a rollback union-find.
#
# BRUTE FORCE (say it first): Rebuild the graph at every
# event time and search it
# For each distinct time, replay all events up to it to
# get the current friend set and run a DFS from person 0
# to see whether everyone is reachable. Simple, correct
# for friend and unfriend events, and the oracle; it
# rebuilds the graph for every time. Time O(T * (N + E))
# for T distinct times. Space O(N + E).


from collections import defaultdict, deque


# BRUTE FORCE (say it first): Rebuild the graph at every
# event time and search it | time O(T * (N + E)) for T
# distinct times
def earliest_connected_rebuild(n, events):
    if n == 1:
        return 0
    for t in sorted({e[0] for e in events}):
        edges = set()
        for et, a, b, kind in sorted(
            events, key=lambda e: e[0]
        ):
            if et > t:
                break
            key = (min(a, b), max(a, b))
            (
                edges.add(key)
                if kind == "friend"
                else edges.discard(key)
            )
        seen, stack = {0}, [0]
        while stack:
            u = stack.pop()
            for a, b in edges:
                for x, y in ((a, b), (b, a)):
                    if x == u and y not in seen:
                        seen.add(y)
                        stack.append(y)
        if len(seen) == n:
            return t
    return -1


# HELPERS for the base question
def by_time(events):
    # events are (time, a, b, "friend" or "unfriend")
    return sorted(events, key=lambda e: e[0])


# FOLLOW-UP 4: Queries ask 'were A and B connected at time
# t?' with no unfriending.
# Answer: Union by size without path compression,
# recording the time of each union on the edge it creates;
# two people are connected at t if the largest time on the
# path between them is at most t.
# NEEDED BY THE BASE QUESTION: it is also the answer to
# the follow-up below
class TimedUnionFind:
    """Union by size, no path compression, so every
    link keeps the time it was made."""

    def __init__(self, n):
        # every person starts alone in a group of one
        self.parent = list(range(n))
        # sizes decide which group hangs under which
        self.size = [1] * n
```

```
        # time a node was linked under its parent
        self.link_time = [0] * n
        # how many separate groups exist right now
        self.groups = n

    def find(self, x):
        # walk up to the leader of x's group
        while self.parent[x] != x:
            x = self.parent[x]
        return x

    def union(self, a, b, t):
        # find the leaders of both groups
        ra, rb = self.find(a), self.find(b)
        # already in one group: nothing changes
        if ra == rb:
            return False
        # make ra the bigger group
        if self.size[ra] < self.size[rb]:
            ra, rb = rb, ra
        # hang the smaller group under the bigger one
        self.parent[rb] = ra
        self.size[ra] += self.size[rb]
        # remember when this link appeared
        self.link_time[rb] = t
        # two groups became one
        self.groups -= 1
        return True

    def connected_at(self, a, b, t):
        # a person is always connected to themselves
        if a == b:
            return True
        # climb from a, noting the newest link
        seen, newest, x = {a: 0}, 0, a
        while self.parent[x] != x:
            newest = max(newest, self.link_time[x])
            x = self.parent[x]
            seen[x] = newest
        # climb from b until we reach an ancestor of a
        newest, y = 0, b
        while y not in seen:
```

```
            # b's leader reached without meeting a
            if self.parent[y] == y:
                return False
            newest = max(newest, self.link_time[y])
            y = self.parent[y]
        # connected at t if all path links are <= t
        return max(newest, seen[y]) <= t


# BASE QUESTION (optimal): earliest time everyone is
# connected (Union-Find on sorted events)
# Idea: Sort by time and union the two people for each
# friend event. The first time the component count reaches
# 1 is the answer; -1 if it never does.
def earliest_all_connected(n, events):
    # zero or one person is connected from the start
    if n <= 1:
        return 0
    uf = TimedUnionFind(n)
    # handle friendships in time order
    for t, a, b, kind in by_time(events):
        # the base question has friend events only
        if kind == "friend":
            uf.union(a, b, t)
            # one group left: everyone is connected now
            if uf.groups == 1:
                return t
    # the groups never merged into one
    return -1
```

```
# FOLLOW-UP 1: Why does unfriending break plain Union-
# Find?
# Answer: A union merges two sets and forgets which edge
# caused it, so removing that edge cannot tell which
# members should separate.
# FOLLOW-UP 2: What is the simple answer if the
# interviewer wants something quick?
# Answer: Rebuild the graph from the events up to time t
# and run BFS or DFS for each query: O(E + N) per query.
# State the cost and then explain how the offline method
# amortizes it.
def connected_with_unfriends(n, events, a, b, t):
    # replay every event up to time t on a set of edges
    edges = set()
```

```
    for time, x, y, kind in by_time(events):
        if time > t:
            break
        pair = (min(x, y), max(x, y))
        # friend adds the edge, unfriend removes it
        if kind == "friend":
            edges.add(pair)
        else:
            edges.discard(pair)
    # build neighbour lists from the surviving edges
    adj = defaultdict(list)
    for x, y in edges:
        adj[x].append(y)
        adj[y].append(x)
    # breadth-first search from a looking for b
    seen, queue = {a}, deque([a])
    while queue:
        u = queue.popleft()
        if u == b:
            return True
        for v in adj[u]:
            if v not in seen:
                seen.add(v)
                queue.append(v)
    return False


# FOLLOW-UP 3: Report how many friend groups exist after
# each event (friends only).
# Answer: Union-find with a component counter: decrement
# when a union merges two different roots; record the
# counter after each event. O(E alpha(N)).
def groups_after_each_event(n, events):
    uf = TimedUnionFind(n)
    counts = []
    # replay friend events in time order
    for t, a, b, kind in by_time(events):
        if kind == "friend":
            uf.union(a, b, t)
            # record the group count after this event
            counts.append(uf.groups)
    return counts


if __name__ == "__main__":
```

```
    # people A, B, C are 0, 1, 2
    ev = [(1, 0, 1, "friend"), (5, 1, 2, "friend")]
    # everyone is connected at time 5
    assert earliest_all_connected(3, ev) == 5
    # a missing friendship means never connected
    assert earliest_all_connected(3, ev[:1]) == -1
    assert earliest_all_connected(1, []) == 0
    # groups: 3 -> 2 -> 1
    assert groups_after_each_event(3, ev) == [2, 1]
    # A and C are apart at time 0, together at time 5
    uf = TimedUnionFind(3)
```

```
    for t, a, b, k in ev:
        uf.union(a, b, t)
    assert not uf.connected_at(0, 2, 4)
    assert uf.connected_at(0, 2, 5)
    # an unfriend breaks the link again
    ev2 = ev + [(7, 1, 2, "unfriend")]
    assert connected_with_unfriends(3, ev2, 0, 2, 6)
    assert not connected_with_unfriends(3, ev2, 0, 2, 8)
```

## NOTES FROM CANDIDATES

The base problem looks innocent; the follow-up about unfriending is the actual signal. In practice, interviewers accept "rebuild graph from scratch per query + BFS" as a working answer if you also articulate why the optimal algorithm is fully dynamic connectivity.

The same family shows up as "friends form a chain over time" or "chess pieces R / L that can move plus walls" — the connectivity backbone is identical.

The follow-up the interviewer is fishing for is consistently a segment tree built over the event timeline; have the offline algorithm ready.

Two algorithmic families converge here: (1) offline — sort all events by time, run union-find, snapshot connectivity at each query T; (2) online with unfriend support — vanilla union-find doesn't handle deletion, so either rebuild per query, use link-cut trees, or apply offline reverse-time processing (treat unfriends as adds in reverse). Clarify with the interviewer which they want.

In an L4 R1 variant, the interviewer requested a graph-and-DFS implementation for the unfriend extension, then continued asking for an optimization that reused the earlier union-find work. Completing the rebuild approach did not end that follow-up.

## PREPARATION

Have weighted/path-compressed union-find ready and warm.

Practice the offline trick: segment tree over time intervals containing each edge, DFS the tree applying / rolling back unions.

Memorize the punchline for fully-dynamic: "online deletion breaks union-find; offline is OK; for online you need Holm–Lichtenberg or link-cut tree (O(polylog n))."